Proof by Mathematical Induction

ANSWER KEY

Section 1: The Principle & Architecture of Induction

1 In a proof by mathematical induction that a proposition $P(n)$ is true for all integers $n \geq 1$, what are the essential logical components?

a. **Base case: prove $P(1)$ is true; Inductive step: prove $P(k) \Rightarrow P(k+1)$; Conclusion by PMI** ✓

b. Verify $P(n)$ for $n = 1, 2, 3, 4, 5$ and deduce the pattern

c. Assume $P(n)$ is true for all n and differentiate both sides

d. Prove $P(k+1) \Rightarrow P(k)$ starting from an arbitrary large integer

The Principle of Mathematical Induction requires establishing the base step (verify $P(1)$), stating the induction hypothesis (assume $P(k)$ is true), showing that $P(k) \Rightarrow P(k+1)$, and stating the formal conclusion.

2 If the base case of an induction proof is verified at $n = 4$, for which set of integers is $P(n)$ established (assuming the inductive step holds for all $k \geq 4$)?

a. $n \in \{4, 5, 6, \ldots\}$ ✓

b. All integers $n \in \mathbb{Z}$

c. $n \in \{1, 2, 3, 4\}$

d. $n \in \mathbb{R}$ with $n \geq 4$

The chain of implications starts at the base case $n = 4$ and propagates to all subsequent integers $n \geq 4$.

3 In proving $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$, what is the algebraic expression for the sum up to $k+1$ using the induction hypothesis?

a. $\frac{k(k+1)}{2} + (k+1)$ ✓

b. $\frac{(k+1)(k+2)}{2} + k$

c. $\frac{k(k+1)}{2} \times (k+1)$

d. $\frac{k(k-1)}{2} + (k+1)$

By definition, $\sum_{r=1}^{k+1} r = \left(\sum_{r=1}^{k} r\right) + (k+1)$. Substituting the inductive hypothesis gives $\frac{k(k+1)}{2} + (k+1)$.

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Section 2: Summation and Series Proofs

4 When proving by induction that $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, factoring $(k+1)$ from $\frac{k(k+1)(2k+1)}{6} + (k+1)^2$ yields:

a. $\frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$ ✓

b. $\frac{k+1}{6}[2k^2+5k+6]$

c. $\frac{(k+1)^2(2k+1)}{6}$

d. $\frac{k+1}{6}[k(2k+1)+1]$

Factor $\frac{k+1}{6}$: $\frac{k+1}{6}[k(2k+1)+6(k+1)] = \frac{k+1}{6}[2k^2+7k+6] = \frac{(k+1)(k+2)(2k+3)}{6}$, which is the target formula at $n = k+1$.

5 For the proposition $\sum_{r=1}^{n} (2r-1) = n^2$, verify the inductive step by simplifying $k^2 + (2(k+1)-1)$.

a. $(k+1)^2$ ✓

b. k^2+2k

c. k^2+2k+3

d. $(k+2)^2$

$k^2 + (2k+2-1) = k^2+2k+1 = (k+1)^2$, completing the induction step.

6 Consider the geometric series $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$. What is the base case check for $n = 1$?

a. $\text{LHS} = 2^0 = 1$, $\text{RHS} = 2^1 - 1 = 1$; **LHS = RHS** ✓

b. $\text{LHS} = 2^1 = 2$, $\text{RHS} = 2^1 - 1 = 1$

c. $\text{LHS} = 0$, $\text{RHS} = 0$

d. $\text{LHS} = 1$, $\text{RHS} = 2$

At $n = 1$, $\text{LHS} = 2^{1-1} = 1$ and $\text{RHS} = 2^1 - 1 = 1$, so the base case is verified.

Section 3: Divisibility and Inequality Proofs

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7 To prove $5^n - 1$ is divisible by 4 for $n \geq 1$, express $5^{k+1} - 1$ in terms of $(5^k - 1)$.

a. $5(5^k - 1) + 4$ ✓

b. $5(5^k - 1) - 4$

c. $4(5^k - 1) + 5$

d. $(5^k - 1) + 5$

$5^{k+1} - 1 = 5 \cdot 5^k - 1 = 5(5^k - 1) + 5 - 1 = 5(5^k - 1) + 4$. If $5^k - 1 = 4M$, this equals $5(4M) + 4 = 4(5M + 1)$, which is divisible by 4.

8 In proving $2^n > n^2$ for $n \geq 5$, assuming $2^k > k^2$, which inequality must be justified to show $2^{k+1} > (k+1)^2$?

a. $2k^2 \geq (k+1)^2 \Leftrightarrow k^2 - 2k - 1 \geq 0$**, which holds for** $k \geq 5$ ✓

b. $k^2 > 2k + 1$ for all $k \in \mathbb{R}$

c. $2^k > 2k$

d. $(k+1)^2 > k^2 + 1$

$2^{k+1} = 2 \cdot 2^k > 2k^2$. We need $2k^2 \geq (k+1)^2 = k^2 + 2k + 1 \Leftrightarrow k^2 - 2k - 1 \geq 0$. For $k \geq 5$, $k(k-2) - 1 \geq 5(3) - 1 = 14 > 0$.

9 To prove $3^{2n} - 1$ is divisible by 8, evaluate the base case $n = 1$.

a. $3^2 - 1 = 8$**, which is divisible by** 8 ✓

b. $3^1 - 1 = 2$

c. $3^0 - 1 = 0$

d. $3^2 - 1 = 9$

At $n = 1$: $3^{2(1)} - 1 = 9 - 1 = 8$, which is divisible by 8 with quotient 1.

Section 4: Calculus and Matrix Applications

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10 Prove by induction that $\frac{d^n}{dx^n}(xe^x) = (x+n)e^x$. What is the inductive step derivative $\frac{d}{dx}[(x+k)e^x]$?

a. $1 \cdot e^x + (x+k)e^x = (x+k+1)e^x$ ✓

b. $(x+k)e^x + e^x = (x+k)e^x$

c. $(x+1)e^x + ke^x$

d. $(x+k+1)xe^x$

By the product rule: $\frac{d}{dx}[(x+k)e^x] = \frac{d}{dx}(x+k) \cdot e^x + (x+k) \cdot e^x = 1 \cdot e^x + (x+k)e^x = (x+k+1)e^x$.

11 If $A = [[1,1],[0,1]]$, prove by induction that $A^n = [[1,n],[0,1]]$. What is the matrix product $A^k \cdot A$?

a. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k+1],[0,1]]$ ✓

b. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,2k],[0,1]]$

c. $[[1,k],[0,1]][[1,1],[0,1]] = [[1,k^2],[0,1]]$

d. $[[1,1],[0,1]]$

Multiplying row by column: $(1)(1)+(k)(0)=1$; $(1)(1)+(k)(1)=k+1$; $(0)(1)+(1)(0)=0$; $(0)(1)+(1)(1)=1$, giving $[[1,k+1],[0,1]]$.

12 A sequence is defined by $u_1 = 5$ and $u_{n+1} = 2u_n - 3$. Prove by induction that $u_n = 2^{n+1} + 3$. Assuming $u_k = 2^{k+1} + 3$, what is u_{k+1}?

a. $2(2^{k+1}+3) - 3 = 2^{k+2} + 6 - 3 = 2^{k+2} + 3$ ✓

b. $2(2^{k+1}) - 3 = 2^{k+2} - 3$

c. $2^{k+1} + 6$

d. $2^{k+2} + 6$

Substitute hypothesis into recurrence:
$u_{k+1} = 2u_k - 3 = 2(2^{k+1}+3) - 3 = 2^{k+2} + 6 - 3 = 2^{(k+1)+1} + 3$, completing the inductive step.

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